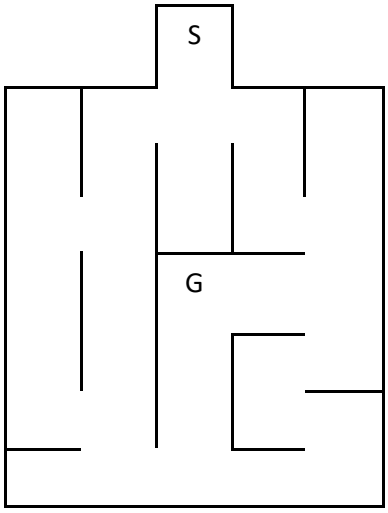
ENN 581 – Semester 2
Week 12 – Mobile robot kinematics

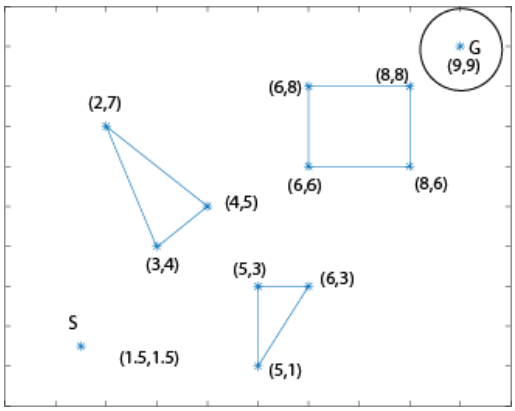
Krishna Manaswi Digumarti

1. Consider the following map with the start S and goal G positions.



   a. Represent this map as a graph. You can solve the problem by hand or use GitHub - petercorke/pgraph-python: Simple graph classes. Refer to the functions add_vertex and add_edge. Or look at the notes from last semester – JupyterNotebook link
   b. Compute the path from S to G using BFS and DFS algorithms.
   c. What do you observe?
   d. How would you upgrade the graph to use A*?

2. Consider a configuration space $\mathcal{C}$ of two dimensions, with each dimension in the range of 0 to 10. The robot starts at point S=(1.5,1.5) and must reach a goal region G=(a circle of radius 1 unit centred at 9,9). Implement an PRM algorithm that finds a path from S to G.



   a. You need a way to keep track of your graph – vertices and edges.
   b. You need a sampling algorithm. Try a random sampler.
   c. You need a collision detection algorithm to determine if a point is inside the obstacle or not. Here is a simple algorithm: https://en.wikipedia.org/wiki/Point_in_polygon
   d. You need a way to determine the nearest point in your graph to the sampled point. You can use Euclidean distance.
   e. You need a way to determine if an edge intersects with a polygon. You can do this by checking if the line interests with every edge of the polygon. You can write a script to do this yourself. Alternatively, check out the shapely library
   f. Check at each step if you reached the goal.
   g. If you reached the goal, print the path that you found.

3. (BONUS) Consider the configuration space map shown above. Write an RRT algorithm that finds a path from S to G. S and G are same as above.